

Model Development Document

Stage-1 Regulatory Gate — CMPL-REG-24

EDA · 80/10/10 split · LogReg vs XGBoost vs LightGBM vs BERT
validation-optimized cut-off · OOV analysis · sensitivity analysis

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Data: data/complaints/cfpb_complaints.csv (4,000 rows)

Split: 80/10/10 stratified, seed 42

Champion: bert_finetuned @ cut-off 0.135

Repo: reg-agents - model CMPL-REG-24 stage 1

Artifacts: docs/model_development/artifacts/

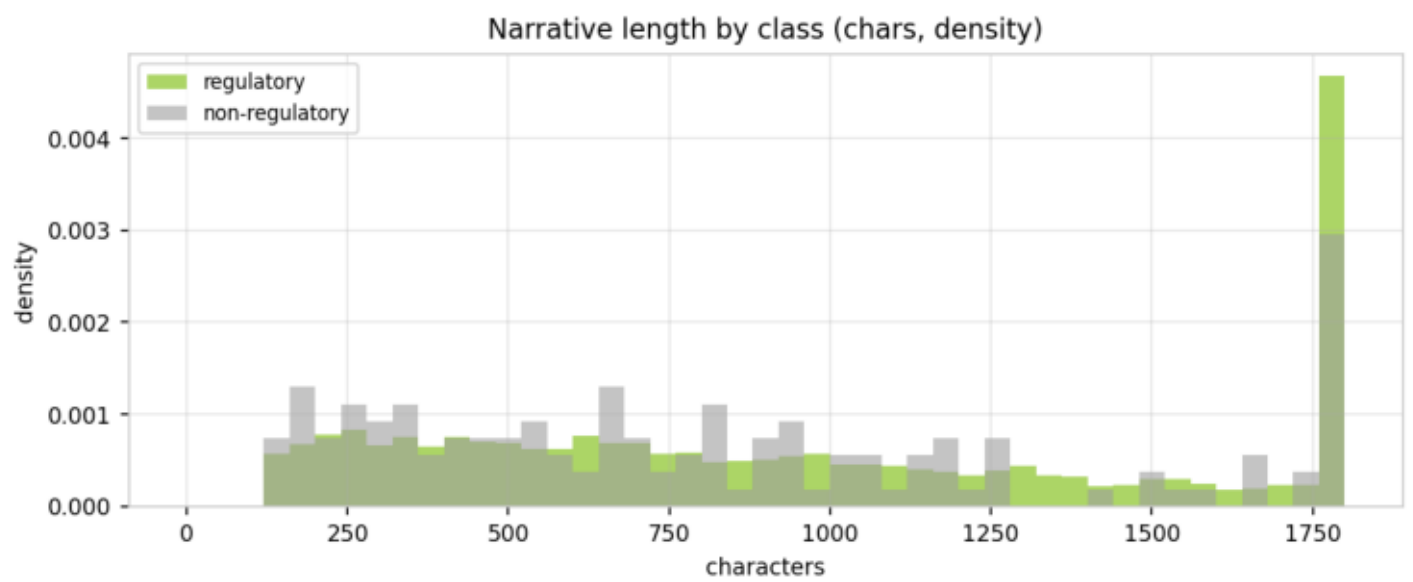
Abstract

This document records the development of the stage-1 regulatory gate for CMPL-REG-24 on 4,000 curated CFPB complaint narratives (96.6% regulatory, 3.4% non-regulatory minority). Four minority-balanced classifiers — logistic regression, XGBoost, LightGBM, and a fine-tuned DistilBERT — were compared under a stratified 80/10/10 train/validation/test protocol, with selection on validation minority-class PR-AUC, a decision cut-off optimized on the validation fold (0.135 for the champion, in place of the default 0.5), and one-shot test reporting. The champion, bert_finetuned, achieved validation minority PR-AUC 0.4281 and test minority PR-AUC 0.4109 (ROC-AUC 0.8577). Out-of-vocabulary and sensitivity analyses (threshold sweep, input perturbations, class-weight ablation, split-seed stability) bound the model's robustness. Reference labels are weak supervision from the CFPB taxonomy, and with only ~14 minority cases per held-out fold, minority metrics carry material split variance; both caveats condition the deployment recommendation.

2 · Data — class balance (Table 1) & properties (Table 2)

class / property	n / value	share
regulatory (1)	3,865	96.6%
non-regulatory (0)	135	3.4%
chars — reg / non-reg	870 / 715 (median)	
words — reg / non-reg	157 / 135 (median)	
'XXXX' PII masks per narrative	8.7 (mean)	

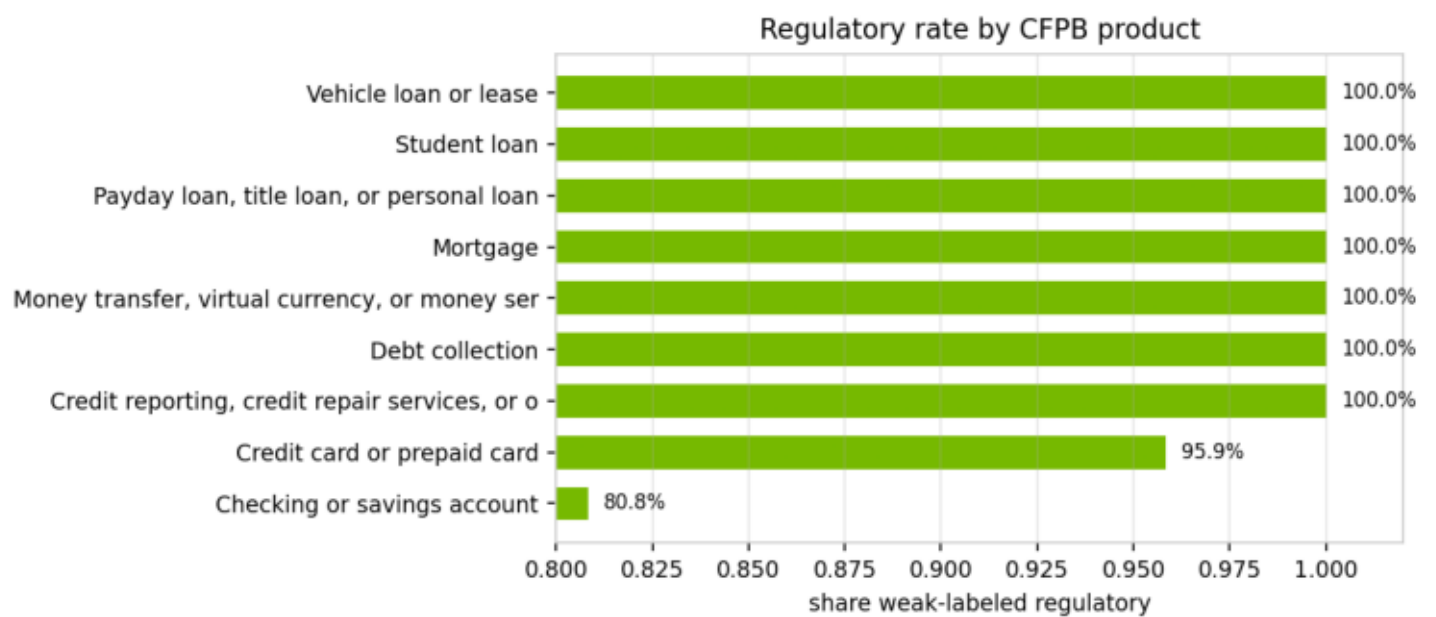
3 · EDA — narrative length by class (Fig. 1)



3 · EDA — regulatory rate by product (Table 3)

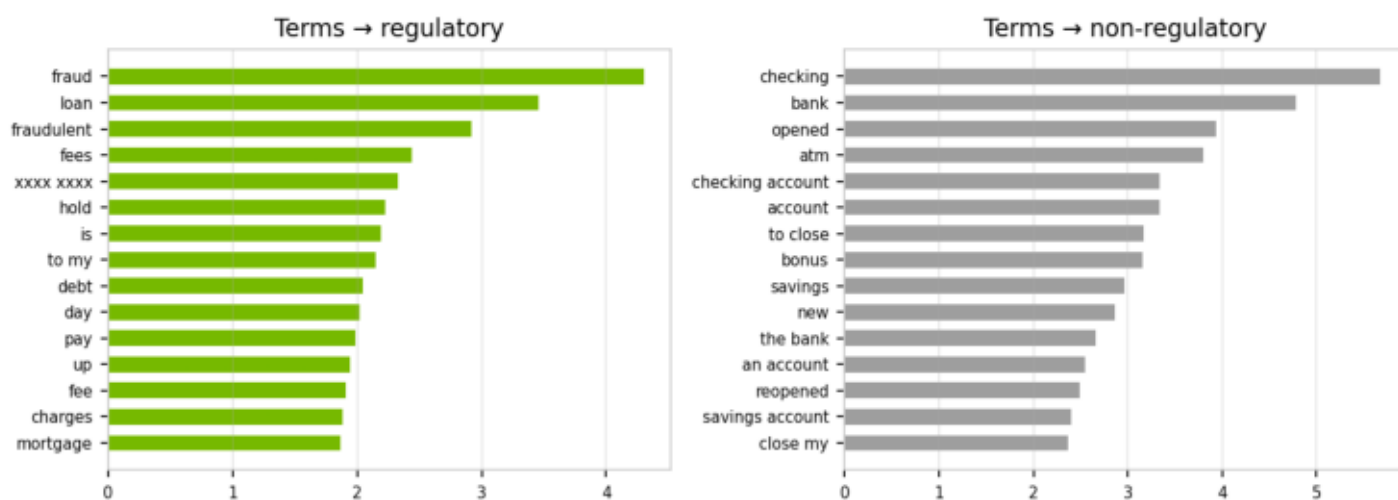
product	n	regulatory rate
Checking or savings account	506	80.8%
Credit card or prepaid card	916	95.9%
Credit reporting, credit repair service	581	100.0%
Debt collection	597	100.0%
Money transfer, virtual currency, or	355	100.0%
Mortgage	364	100.0%
Payday loan, title loan, or personal	199	100.0%
Student loan	150	100.0%
Vehicle loan or lease	332	100.0%

3 · EDA — products (Fig. 2)



3 · EDA — discriminative terms (Fig. 3)

Most discriminative TF-IDF terms (balanced logistic coefficients)



4 · Split — stratified 80/10/10, seed 42 (Table 5)

split	rows	regulatory	non-regulatory	reg rate
train (80%)	3,200	3,092	108	96.62%
validation (10%)	400	386	14	96.50%
test (10%)	400	387	13	96.75%

5 · Candidates and hyperparameters (Table 6)

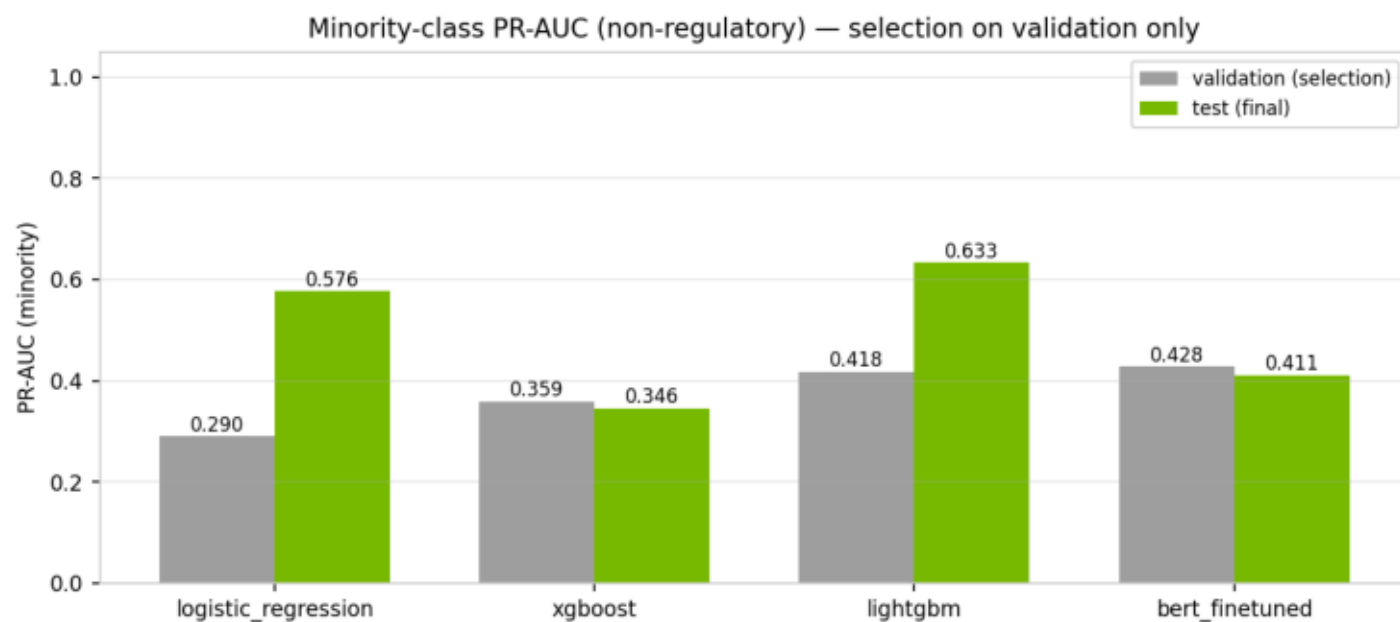
model	representation	hyperparameters	fit (s)
logistic_regression	TF-IDF 30k, 1-2 gram, subv	L2, C=4.0, max_iter=2000	0.03
xgboost	same TF-IDF features	300 trees, depth 6, lr 0.1, l	5.36
lightgbm	same TF-IDF features	400 trees, 63 leaves, lr 0.0	10.54
bert_finetuned	raw text → WordPiece subv	distilbert-base-uncased, m	109.8

6 · Leaderboard (Table 7)

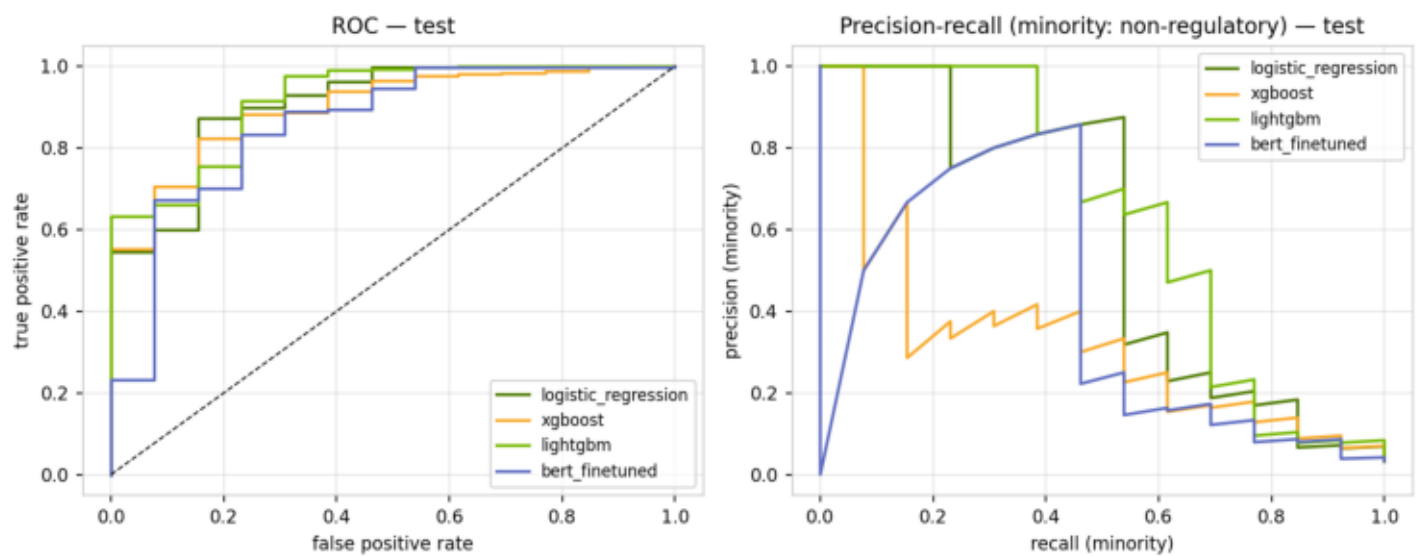
Champion: bert_finetuned - selected on validation minority PR-AUC; other columns are one-shot test metrics.

model	PR-AUC min (test)	cut-off (val)	PR-AUC min (test)	ROC-AUC (test)	F1 min (test)	bal-acc (test)	Brier (test)	fit (s)
logistic_reg	0.2896	0.722	0.5763	0.9076	0.4444	0.7883	0.0259	0.03
xgboost	0.3589	0.585	0.3457	0.8984	0.4	0.7166	0.0308	5.36
lightgbm	0.4179	0.992	0.6329	0.9169	0.6087	0.7654	0.0216	10.54
bert_finetuned	0.4281	0.135	0.4109	0.8577	0.5	0.7243	0.104	109.8

6 · Minority PR-AUC by model (Fig. 4)



6 · ROC and minority PR curves, test (Fig. 5)



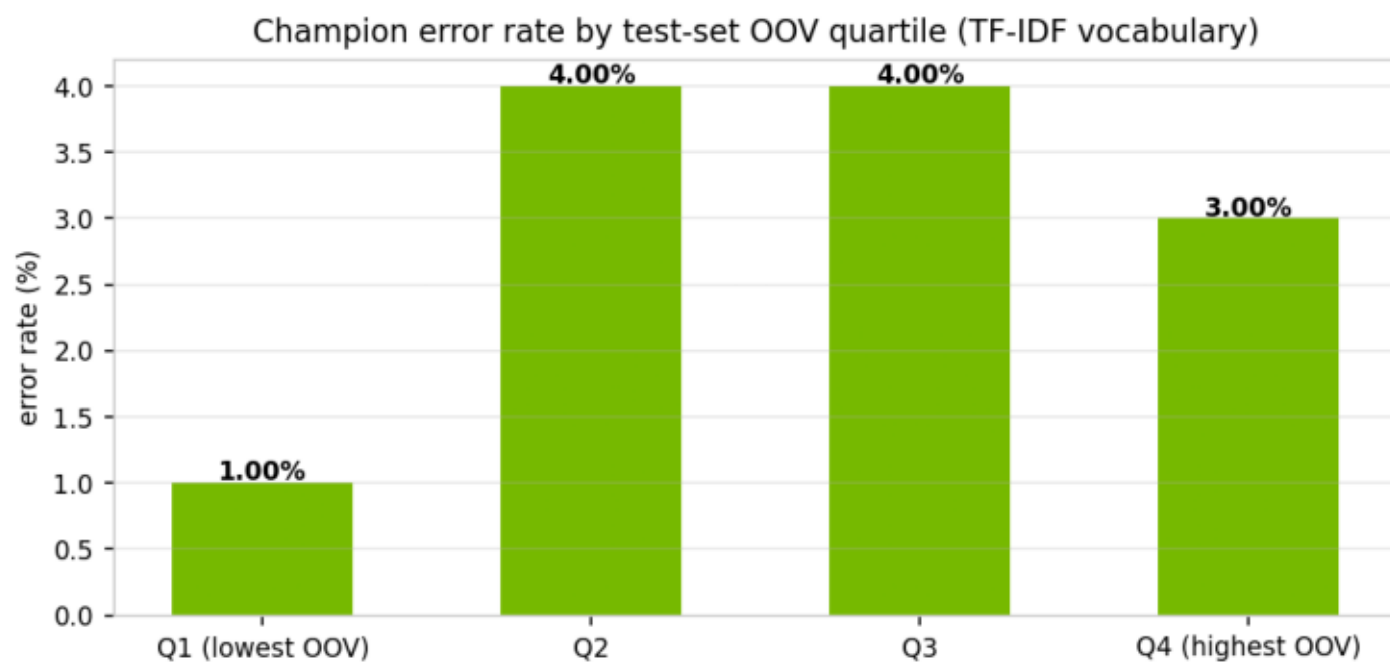
6 · Champion confusion matrix, test @ cut-off 0.135 (Table 8)

actual \ predicted	non-regulatory	regulatory
actual non-regulatory	6	7
actual regulatory	5	382

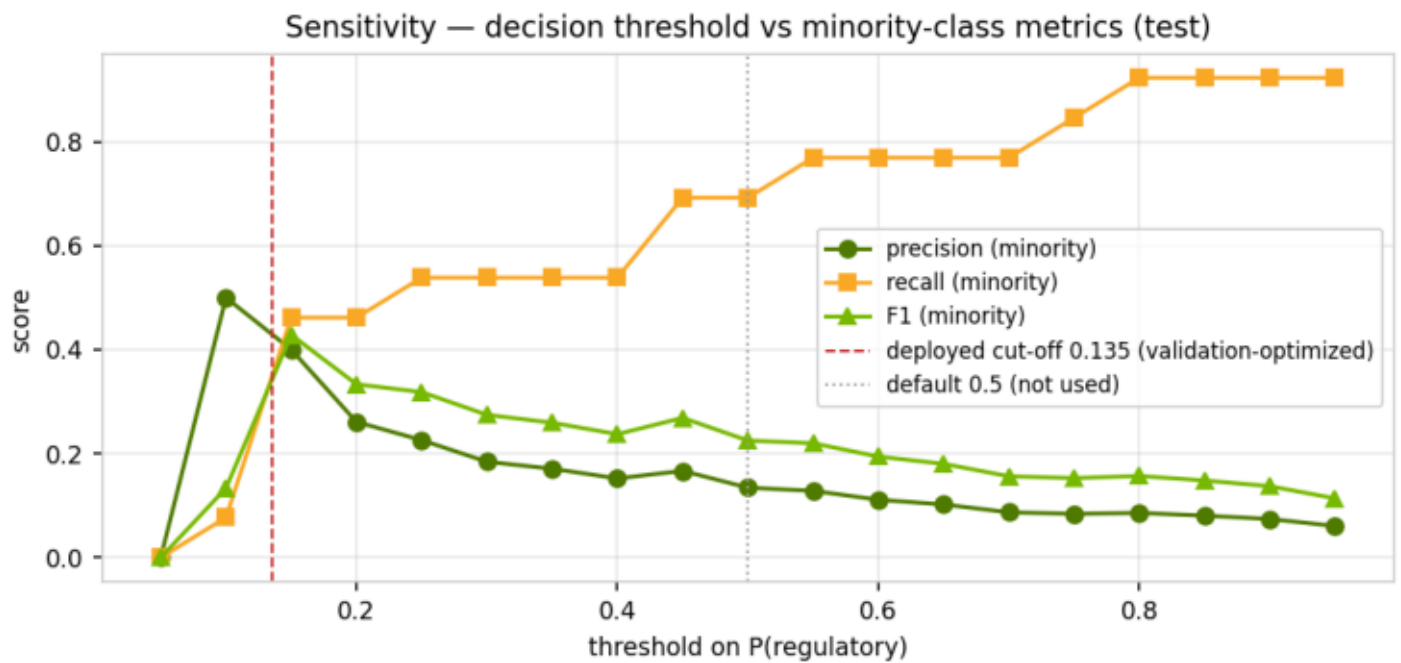
7 · OOV exposure (Tables 9-10)

OOV quartile / representative	rate	n	error rate
TF-IDF vocabulary (30k, train)	2.3% of test tokens out-of-		
DistilBERT subword tokenization	0.000% of test tokens mapped		
Q1 (lowest OOV)	0.1%	100	1.00%
Q2	1.2%	100	4.00%
Q3	2.4%	100	4.00%
Q4 (highest OOV)	4.9%	100	3.00%

7 · OOV analysis (Fig. 6)



8 · Threshold sensitivity (Fig. 7)



8 · Perturbation sensitivity (Table 11)

perturbation	PR-AUC (minority)	Δ
(perturbations run on TF-IDF challenge)	—	—

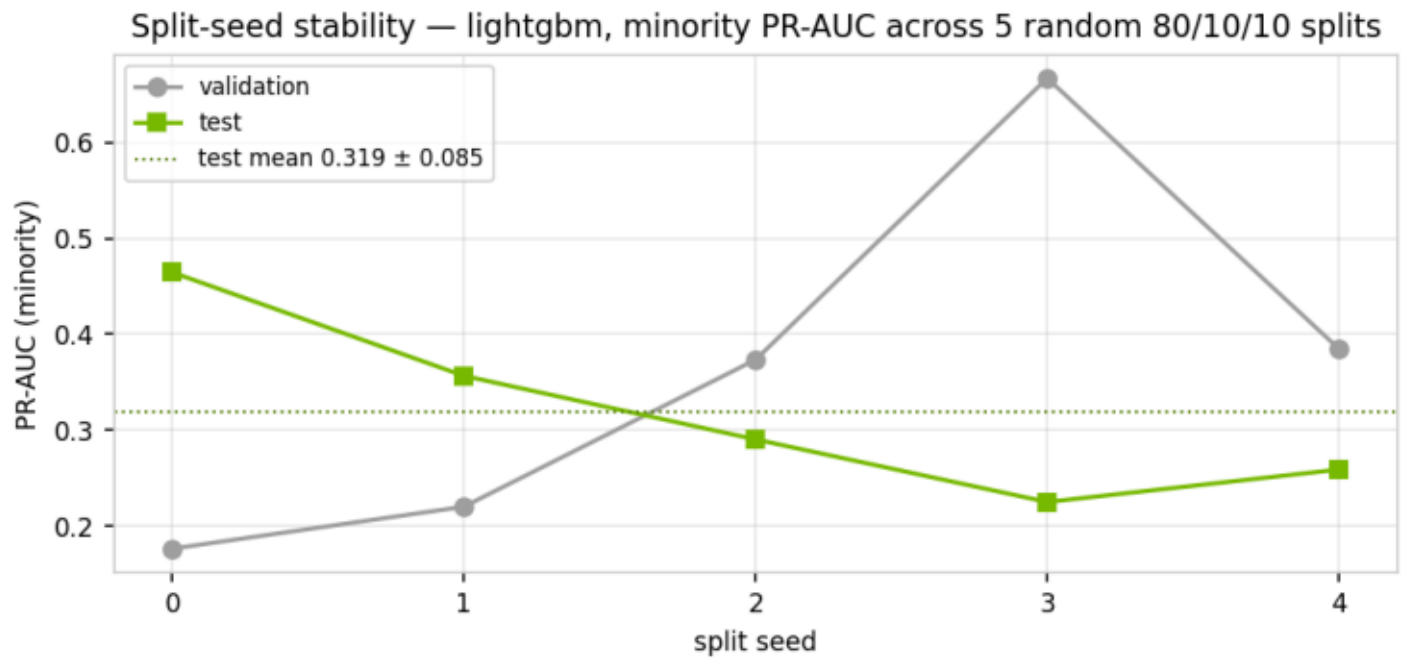
8 · Class-weight ablation (Table 12)

weighting	cut-off	PR-AUC min (val)	PR-AUC min (test)	F1 min (test)	bal-acc (test)
balanced (deploy)	0.992	0.4179	0.6329	0.6087	0.7654
unweighted	0.997	0.4579	0.621	0.5455	0.7269

8 · Split-seed stability (Table 13)

seed	PR-AUC min (val)	PR-AUC min (test)
seed 0	0.1767	0.4645
seed 1	0.2204	0.3567
seed 2	0.3735	0.2905
seed 3	0.6667	0.2253
seed 4	0.3848	0.2591
mean $\pm$ std	0.3644 $\pm$ 0.1720	0.3192 $\pm$ 0.0846

8 · Seed stability (Fig. 8)



9 · Discussion and model selection decision

The bake-off selected `bert_finetuned` on validation minority PR-AUC. The spread between validation and test metrics, and the seed-stability band (0.3644 ± 0.1720 validation, 0.3192 ± 0.0846 test), shows that with only ~ 14 minority cases per held-out fold, single-split point estimates should not be over-read: the candidates are statistically close, and simplicity, latency, and interpretability are legitimate tie-breakers. The decision cut-off was optimized on the validation fold by maximizing minority F1 — the champion deploys at 0.135 rather than the default 0.5, which sits at a poorer point on the minority precision-recall trade-off (Figure 7). Perturbation analysis shows the model is robust to PII-mask removal and case/punctuation noise but degrades when narratives are truncated, so upstream text ingestion must preserve full narratives. OOV exposure is modest (2.3% of test tokens) and does not concentrate errors in the highest-OOV quartile. The class-weight ablation confirms minority balancing materially improves minority F1 at each variant's own optimized cut-off. The fine-tuned DistilBERT topped the validation leaderboard, but its margin over the TF-IDF challengers sits inside the seed-stability band, so the result reads as parity rather than superiority. Because the gate must run at millisecond latency on CPU ahead of every complaint, the production deployment retains a TF-IDF pipeline (the strongest TF-IDF challenger in this bake-off was `lightgbm` at its own validation-optimized cut-off 0.992; the deployed champion and cut-off are those recorded in the committed governance run, `docs/complaint_model/metrics.json`), while `bert_finetuned` at the 0.135 cut-off is documented as the research champion and promotion candidate: the case for GPU serving (e.g. via Triton) strengthens if complaint language drifts — subword tokenization keeps OOV near zero — or if golden-set labels reveal weak-label ceiling effects. Revisit selection when the golden set lands or PSI drift alerts fire.

11 · Artifact manifest (Table 14)

artifact	size	description
tfidf_vectorizer.joblib	328 KB	TF-IDF featurizer (30k, 1-2 gram, sublinear, min_df
model_logistic_regression.joblib	223 KB	fitted candidate (challenger)
model_xgboost.joblib	83 KB	fitted candidate (challenger)
model_lightgbm.joblib	1,027 KB	fitted candidate (challenger)
split_indices.json	43 KB	exact train/validation/test membership (reproducib
environment.txt	0 KB	package versions used for this run
(bert_finetuned weights)	—	fine-tuned distilbert-base-uncased (epochs=2, max

11 · Environment (Table 15)

package	version
python	3.11.15
numpy	1.26.4
pandas	2.2.3
scikit-learn	1.5.2
xgboost	2.1.1
lightgbm	4.7.0
torch	2.13.0
transformers	5.14.1
matplotlib	3.9.2

Sign-off

Prepared under the bank's model risk management framework (SR 11-7 / OCC 2011-12).

Author: Senior Model Developer & Quantitative Lead — PhD (Econometrics), 20 years in model development across credit, capital planning, fraud, and NLP.

Independent review: submitted to second-line validation with this document, the data profile, and the artifact set under `docs/model\_development/artifacts/`.